

2.3.2 Limits at Infinity

In this section we would like to be able to calculate limits where x is approaching infinity or negative infinity. While the rigorous definitions of what this means can be given, we will stick with intuitive definitions throughout.

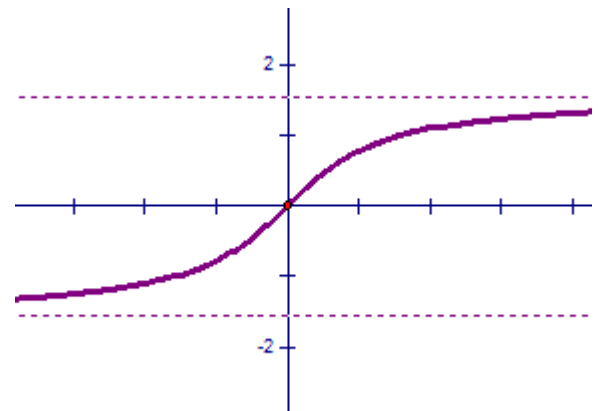
Motivation:

What would you expect the following two limits to be: $\lim_{x \rightarrow \infty} \arctan(x)$ and $\lim_{x \rightarrow -\infty} \arctan(x)$?

By looking at some values of x , and observing the graph (see below), one starts to suspect that:

$$\lim_{x \rightarrow \infty} \arctan(x) = \frac{\pi}{2} \text{ and } \lim_{x \rightarrow -\infty} \arctan(x) = -\frac{\pi}{2}.$$

Note: The graph of $\arctan$ seems to have two different horizontal asymptotes, that is, $y = \frac{\pi}{2}$ as x goes to infinity and $y = -\frac{\pi}{2}$ as x goes to negative infinity. This makes us suspect that horizontal asymptotes are related to the limits when x goes to infinity or negative infinity.



Definition (Intuitive): Let f be a function defined on some interval (a, ∞) . Then, $\lim_{x \rightarrow \infty} f(x) = L$ means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.

Note: One can give an intuitive definition of $\lim_{x \rightarrow \infty} f(x) = L$ by replacing “sufficiently large” with “sufficiently small”.

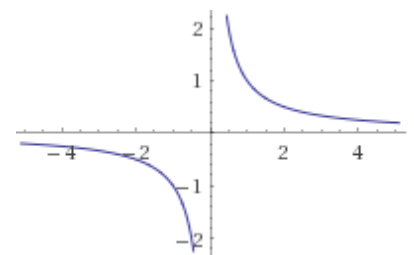
Note: We will also consider the cases where the function goes to positive/negative infinity as x goes to positive/negative infinity later in this lecture.

Example 1: Find $\lim_{x \rightarrow \infty} \frac{1}{x}$ and $\lim_{x \rightarrow -\infty} \frac{1}{x}$.

First, note that the graph of $f(x) = \frac{1}{x}$ pictured to the right.

We also note that 1 divided by a large positive number is close to 0, and 1 divided by a very small number is close to 0. So, we conclude

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0 \text{ and } \lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$



Note: From the graph there appears to be a horizontal asymptote of $y = 0$. We have the following definition.

Definition: The line $y = L$ is called a horizontal asymptote of the curve $y = f(x)$ if either $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.

Note: It is important to understand that a function's graph may cross its horizontal asymptote. All that is important when identifying horizontal asymptotes is the limit.

Note: By the above definition we have that $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$ are horizontal asymptotes of the graph of the inverse tangent function.

Example 2: Find each limit. For limits where x approaches infinity or negative infinity identify any horizontal asymptote(s) of the graph of the corresponding function.

a) $\lim_{x \rightarrow -\infty} \frac{1}{x^2}$

We have $\lim_{x \rightarrow -\infty} \frac{1}{x^2} = 0$, and the graph of $f(x) = \frac{1}{x^2}$ has a horizontal asymptote of $y = 0$.

b) $\lim_{x \rightarrow \infty} \sin(x)$

Since sine oscillates between 1 and -1, this limit does not approach any single number. So, the limit DNE, and the graph of $f(x) = \sin(x)$ has no horizontal asymptotes.

c) $\lim_{x \rightarrow -\infty} e^x$

Recall that $e^{-a} = \frac{1}{e^a}$. So, $\lim_{x \rightarrow -\infty} e^x = 0$, and the graph of $f(x) = e^x$ has a horizontal asymptote of $y = 0$.

d) $\lim_{x \rightarrow 2^+} \arctan\left(\frac{1}{x-2}\right)$

We have $\lim_{x \rightarrow 2^+} \arctan\left(\frac{1}{x-2}\right) = \lim_{t \rightarrow \infty} \arctan(t) = \frac{\pi}{2}$.

e) $\lim_{x \rightarrow \infty} x^2 - 20x + 10$

We have that $\lim_{x \rightarrow \infty} x^2 - 20x + 10 = \infty$. Therefore the limit DNE, and the graph of $f(x) = x^2 - 20x + 10$ has no horizontal asymptotes.

Part (e) of this example leads us to the following observation.

Fact: When we are computing the limit of a polynomial as x goes to infinity or negative infinity, the highest power term dominates.

Example 3: Find $\lim_{x \rightarrow -\infty} (7x^3 - x^2 + 20x)$

We calculate: $\lim_{x \rightarrow -\infty} (7x^3 - x^2 + 20x) = \lim_{x \rightarrow -\infty} 7x^3 = -\infty$.

Part (a) of example 2 leads us to the following theorem.

Theorem: If $r > 0$ is a rational number, then $\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$. Moreover, if x^r is defined for all x , $\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$.

This theorem is useful for calculating limits of rational functions and functions close to rational functions when x goes to infinity or negative infinity.

Example 4: Find $\lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5}$

We would like to state $\lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5} = \lim_{x \rightarrow -\infty} \frac{4x^2}{2x^3} = \frac{\infty}{\infty}$, however $\frac{\infty}{\infty}$ is not a well-defined mathematical value (we will discuss forms like this in more detail in section 4.8). We cannot stop here but instead will make use of the previous theorem.

Our strategy is to strategically multiply by one so as to get expressions that look like the expressions in the above theorem. To do this, we divide the numerator and denominator by the largest power of x present in the denominator (to avoid dividing by 0). Specifically, we calculate:

$$\lim_{x \rightarrow -\infty} \frac{(4x^2 - x)}{(2x^3 - 5)} \cdot \frac{1/x^3}{1/x^3} = \lim_{x \rightarrow -\infty} \frac{\frac{4}{x} - \frac{1}{x^2}}{2 - \frac{5}{x^3}} = \frac{0 - 0}{2 - 0} = 0.$$

Example 5: Find each of the following limits.

a) $\lim_{x \rightarrow \infty} \frac{3x + 5}{6x - 8}$

$$\lim_{x \rightarrow \infty} \frac{3x + 5}{6x - 8} = \lim_{x \rightarrow \infty} \frac{3x + 5}{6x - 8} \cdot \frac{1/x}{1/x} = \lim_{x \rightarrow \infty} \frac{3 + \frac{5}{x}}{6 - \frac{8}{x}} = \frac{3 + 0}{6 - 0} = \frac{3}{6} = \frac{1}{2}$$

So we can say that $y = \frac{1}{2}$ is a horizontal asymptote on the graph of $f(x) = \frac{3x + 5}{6x - 8}$.

$$\text{b) } \lim_{x \rightarrow -\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x}$$

$$\lim_{x \rightarrow -\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x} = \lim_{x \rightarrow -\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x} \cdot \frac{1/x}{1/x} = \lim_{x \rightarrow -\infty} \frac{5x^2 - 2x + \frac{1}{x}}{\frac{1}{x} - 3} = \lim_{x \rightarrow -\infty} \frac{5x^2 - 2x + 0}{0 - 3} = \lim_{x \rightarrow -\infty} \frac{5x^2}{-3} = \frac{\infty}{-3} = -\infty$$

So we have found there is no horizontal asymptote as $x \rightarrow -\infty$ on the graph of $f(x) = \frac{5x^3 - 2x^2 + 1}{1 - 3x}$.

Example 6: Find each of the following limits.

$$\text{a) } \lim_{x \rightarrow \infty} \frac{\sqrt{2x^2 + 1}}{3x - 5}$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{2x^2 + 1}}{3x - 5} = \lim_{x \rightarrow \infty} \frac{\sqrt{2x^2 + 1}}{(3x - 5)} \cdot \frac{1/x}{1/x} = \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{2x^2 + 1}}{x}}{3 - \frac{5}{x}}$$

Note that since $x \rightarrow \infty$ we can assume that $x > 0$ and therefore we can say that $x = \sqrt{x^2}$.

$$\text{So we have } \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{2x^2 + 1}}{x}}{3 - \frac{5}{x}} = \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{2x^2 + 1}}{\sqrt{x^2}}}{3 - \frac{5}{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{2x^2 + 1}{x^2}}}{3 - \frac{5}{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{2 + 0}}{3 - 0} = \frac{\sqrt{2}}{3}$$

$$\text{b) } \lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 1}}{3x - 5}$$

This limit is more challenging than part (a). The subtle fact to understand is that since $x \rightarrow -\infty$ we can assume that $x < 0$ and therefore we can say that $x = -\sqrt{x^2}$. Now, we calculate:

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 1}}{(3x - 5)} \cdot \frac{1/x}{1/x} = \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{2x^2 + 1}}{x}}{3 - \frac{5}{x}} = \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{2x^2 + 1}}{-\sqrt{x^2}}}{3 - \frac{5}{x}} = \frac{-\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}} = \frac{-\sqrt{2}}{3}.$$

Question: What do the answers to the most recent example imply about the horizontal asymptotes of

$$g(x) = \frac{\sqrt{2x^2 + 1}}{3x - 5}?$$

Example 7: Find $\lim_{x \rightarrow \infty} \sqrt{x^6 + 5x^3} - x^3$. (Hint: Attempt to multiply by 1 in the form of the conjugate)

$$\begin{aligned} \lim_{x \rightarrow \infty} \sqrt{x^6 + 5x^3} - x^3 &= \lim_{x \rightarrow \infty} \frac{\sqrt{x^6 + 5x^3} - x^3}{1} \cdot \frac{\sqrt{x^6 + 5x^3} + x^3}{\sqrt{x^6 + 5x^3} + x^3} = \lim_{x \rightarrow \infty} \frac{x^6 + 5x^3 - x^6}{\sqrt{x^6 + 5x^3} + x^3} = \lim_{x \rightarrow \infty} \frac{5x^3}{\sqrt{x^6 + 5x^3} + x^3} \\ &= \lim_{x \rightarrow \infty} \frac{5x^3}{\sqrt{x^6 + 5x^3} + x^3} \cdot \frac{1/x^3}{1/x^3} = \lim_{x \rightarrow \infty} \frac{5}{\frac{\sqrt{x^6 + 5x^3}}{x^3} + 1} \end{aligned}$$

Note that since $x \rightarrow \infty$ we can assume that $x > 0$ and therefore $x^3 = \sqrt{x^6}$. So we have

$$\lim_{x \rightarrow \infty} \frac{5}{\frac{\sqrt{x^6 + 5x^3}}{x^3} + 1} = \lim_{x \rightarrow \infty} \frac{5}{\frac{\sqrt{x^6 + 5x^3}}{\sqrt{x^6}} + 1} = \lim_{x \rightarrow \infty} \frac{5}{\sqrt{1 + \frac{5}{x^3}} + 1} = \frac{5}{\sqrt{1+0} + 1} = \frac{5}{2}.$$

Example 8:

(a) Find $\lim_{x \rightarrow \infty} e^x$.

We note that the exponential function gets arbitrarily large as x gets arbitrarily large. So, $\lim_{x \rightarrow \infty} e^x = \infty$

(b) Suppose that $f(x) = (1-x)(x+1)^3(x-4)^3$. Find the limit of this function as x goes to infinity, and find its limit as x goes to negative infinity.

Note that f is a polynomial with leading term: $-x^7$. So, $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} (-x^7) = -\infty$ and

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} (-x^7) = \infty.$$

(c) Use limits to find all the horizontal asymptotes of the graph of $f(x) = \frac{5e^x + 6}{3 + e^x}$.

We note that $\lim_{x \rightarrow -\infty} \frac{5e^x + 6}{3 + e^x} = \frac{5(0) + 6}{3 + 0} = 2$ and $\lim_{x \rightarrow \infty} \frac{(5e^x + 6)}{(3 + e^x)} \cdot \frac{1/e^x}{1/e^x} = \lim_{x \rightarrow \infty} \frac{5 + \frac{6}{e^x}}{\frac{3}{e^x} + 1} = \frac{5 + 0}{0 + 1} = 5$. So, the graph of

the function has horizontal asymptotes of $y = 2$ and $y = 5$.